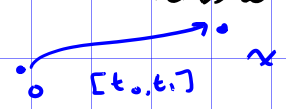
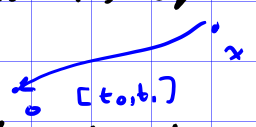
Complete Controllability

Definitions

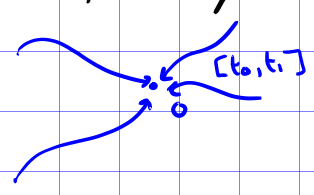
Reachable: $x \in \mathbb{R}^n$ is reachible on $[t_0, t_1]$ if $\exists\ u \in \mathcal{U}$ s.t. with $x(t_0) = 0$ we have $x(t_1) = x$



Controllable: $x \in \mathbb{R}^n$ is controllable on $[t_0, t_1]$ if $\exists\ u \in \mathcal{U}$ s.t. with $x(t_0) = x$ we have $x(t_1) = 0$



Completely Controllable: The system is completely controllable on $[t_0, t_1]$ if every state $x \in \mathbb{R}^n$ is controllable on $[t_0, t_1]$



Relate Reachable and Controllable

I) x is reachible iff

$$x = 0 + \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) u(\tau)\, d\tau \quad \text{for some } u(\tau)$$

$$\iff x \in \mathcal{R}[t_0, t_1] \quad \text{and we proved}$$

$$\iff x \in \mathrm{Im}(W_R(t_0, t_1))$$

II) x is controllable iff

$$0 = \Phi(t_1, t_0) x + \int_{t_0}^{t_1} \Phi(t, \tau) B(\tau) u(\tau)$$

$$\iff x = \mathcal{C}[t_0, t_1]$$

$$\iff -\Phi(t_1, t_0) x \in \mathcal{R}[t_0, t_1] \iff -\Phi(t_1, t_0) \in \mathrm{Im}(W_R(t_0, t_1))$$

$$\iff \Phi(t_1, t_0) x \in \mathrm{Im}(W_R(t_0, t_1))$$

↳ If V is a subspace $v \in V$ then $-v \in V$

Relating CC to Reachable

Proposition: The system is completely controllable (CC) on $[t_0\ t_1]$ iff every state is reachible on $[t_0, t_1]$

I) Proof

$CC \iff \Phi(t_1, t_0)x \in \mathrm{Im}(W_R(t_0, t_1)) \ \forall\, x \in \mathbb{R}^n$

$\iff \mathrm{Im}(\Phi(t_1, t_0)) = \mathbb{R}^n$

So iff C.C. $x' = \Phi(t_1, t_0)x \in \mathrm{Im}(W_R(t_0, t_1)) \ \forall\, x' \in \mathbb{R}^n$

$\iff$ Every state is reachible

$\hookrightarrow x' \in \mathrm{Im}(W_R(t_0, t_1)) \ \forall\, x' \in \mathbb{R}^n$

Moreover, if this is to happen then $W_R(t_0, t_1)$ is non-singular

II) Theorem

C.C. $\iff W_R(t_0, t_0)$ is non-singular

True because

LTI Systems - Motivational Example

Discrete Time: Is x reachible from $x_0 = 0$ in N steps

$N < n$ & $x_{k+1} = Ax_k + Bu_k$

$x_1 = Bu_0$

$x_2 = Ax_1 + Bu_1 = ABu_0 + Bu_1 = [B \ AB] \begin{bmatrix} u_1 \\ u_0 \end{bmatrix}$

$x_3 = Ax_2 + Bu = AABu_1 + Bu_2 \quad [B \ AB \ A^2B] \begin{bmatrix} u_2 \\ u_1 \\ u_0 \end{bmatrix}$

$\vdots$

$x_N = [B \ AB \ A^2B \cdots A^{N-1}B] \begin{bmatrix} u_{n-1} \\ \vdots \\ u_0 \end{bmatrix}$

When does $x_N = x$ have a solution? $x \in \mathrm{Im}([B \ AB \cdots {}^{N-1}B])$

I) Definition: The rank of M ($[A \ AB \cdots]$) is the max # of linearly independent rows

If $\mathrm{rank}(M) = n$ then M is full rank

Note: If $\mathrm{rank}(M) = n$ then MM^T is invertible (Used later)

II) What is rank(M)?

Cayley-Hamilton Theorem allows us to say:

$$A^n B = \alpha_0 B + \alpha_1 AB + \cdots + \alpha_{n-1} A^{n-1} B$$

$\Rightarrow$ The rows $A^n B$ are linearly dependent on the rows of $[B \;\; AB \;\; \cdots \;\; A^{n-1} B]$

Similarly $A^k B = \beta_0 B + \cdots \beta_{n-1} A^{n-1} B$ for any $k \geq 0$

A) Theorem:

$$\text{rank}[A \;\; AB \cdots A^{N-1} B] = \text{rank}([B \;\; AB \;\; \cdots \;\; A^{n-1} B])$$

for $N \geq n$

$\Rightarrow$ All we need to do is check rank($\mathcal{M}$) where $\mathcal{M} =$

$\mathcal{M}$ is known as Reachability/Controllability Matrix

Furthermore

$$\text{Im}(W_R(t_0, t_1)) = \text{Im}(\mathcal{M}) \quad \text{for continuous systems}$$

B) Theorem

$$R[t_0, t_1] = \text{Im}(W_R(t_0, t_1)) = \text{Im}(\mathcal{M}) = \text{Im}(W_c(t_0, t_1) = \mathcal{C}[t_0, t_1]$$

1) Proof

i) $x \in \text{Im}(W_R(t_0, t_1)) \Rightarrow x \in \text{Im}(\mathcal{M})$

$$\Rightarrow \exists\, u \text{ s.t. } x_1 = \int_{t_0}^{t_1} e^{A(t_1 - \tau)A} B u(\tau)\, d\tau$$

Recall $e^{tA} = \sum_{i=0}^{n-1} \alpha_i(t) A^i \;\; \forall\, t \in [t_0, t_1]$

$$\Rightarrow x_1 = \int_{t_0}^{t_1} \sum_{i=0}^{n-1} \alpha_i(t_1 - \tau) A^i B u(\tau)\, d\tau$$

$$= \int_{t_0}^{t_1} \sum_{i=0}^{n-1} A^i B u(\tau) \alpha_i(t_1 - \tau)\, d\tau$$

The summation
$a_1b_1 + a_2b_2 + a_3b_3$
$= a^T b$

$$= \sum_{i=0}^{n-1} A^i B \int_{t_0}^{t_1} u(\tau)\, \alpha_i(t_1-\tau)\, d\tau$$

$$= \underbrace{[B \;\; AB \;\; \cdots \;\; A^{n-1}B]}_{M} \underbrace{\begin{bmatrix} \int_{t_0}^{t_1} u(\tau)\alpha_0(t_1-\tau)\,d\tau \\ \vdots \\ \int_{t_0}^{t_1} u(\tau)\alpha_{n-1}(t_1-\tau)\,d\tau \end{bmatrix}}_{\eta_1}$$

$$\Rightarrow x_1 = M\eta_1$$

$$\Rightarrow x_1 \in \mathrm{Im}(M)$$

ii) $x \in \mathrm{Im}(M) \Rightarrow x \in \mathrm{Im}(W_R(t_0,t_1))$

if $x \in \mathrm{Im}(M) \Rightarrow \exists\, v$ s.t. $x_1 = Mv$ (Definition of Image)

Suppose $\eta_1 \in N(W_R(t_0,t_1))$

↳ We know $B^T(\tau)\Phi^T(t_1,\tau)\eta_1 = 0$

Now suppose: $B^T(e^{(t_1-\tau)A})^T \eta_1 = 0$ or $\eta_1^T e^{(t_1-\tau)A} B = 0$

$$\frac{d}{d\tau}\left(\eta_1^T e^{(t_1-\tau)A} B\right) = -\eta_1^T A e^{(t_1-\tau)} B = 0$$

$$\frac{d^2}{d\tau^2}(\text{"} \qquad \text{"}) = \eta_1^T A^2 e^{(t_1-\tau)} B = 0$$

$$\vdots$$

$$\frac{d^k}{d\tau^k}(\text{"} \qquad \text{"}) = -(1)^k \eta_1^T A^k e^{(t_1-\tau)A} B = 0$$

For $\tau = t_1 \Rightarrow (-1)^k \eta_1^T A^k \underbrace{e^{(t_1-t_1)A}}_{I} B = 0$

$$\Rightarrow (-1)^k \eta_1^T A^k B = 0$$

$$\Rightarrow \eta_1^T A^k B = 0 \quad \forall\, k \geq 0$$

$$\eta_1^T [B \;\; AB \;\cdots\; A^{n-1}B] = 0$$

$$\eta_1^T B = 0, \; \eta_1^T AB = 0, \; \cdots$$

Now to actually prove this:

$$x_1 = \Gamma v \quad , \quad \eta_1^T x_1 = \eta_1^T \Gamma v = 0 \quad \text{if } \eta_1 \in N(W_R(t_0,t_1))$$

$$\Rightarrow x_1 \in \left(N(W_R(t_0,t_1)) \right)^{\perp} = \mathrm{Im}(W_R(t_0,t_1))$$

$$\Leftrightarrow x_1 \in \mathrm{Im}(W_R(t_0,t_1))$$

LTI Reversibility and Time Scaling

I) Time Reversibility

A) Continuous: Controllable and Reachable subspaces coincide due to the fact that $\Phi(t_1,\tau) = e^{A(t-\tau)}$ is full rank (inverse exists)

B) Discrete

Must consider reachable and controllable separately unless A is nonsingular

II) Time Scaling

A) Continuous

- Subspaces do not depend upon the time interval
- Implication: If you can travel between x_0 & x_1 in $[t_0,t_1]$ you can do it in any interval

B) Discrete

- Subspaces depend on the interval if # steps $< n$
- If you can traverse between x_0 & x_1 then you can do it in n steps.

Summary

x is controllable: $\Phi(t_1,t_0)\, x \in \mathrm{Im}(W_R(t_0,t_1))$

x is reachible: $x \in \mathrm{Im}(W_R(t_0,t_1))$

System is completely controlable

$$x \in \mathrm{Im}(W_R(t_0,t_1)) \ \forall x \in \mathbb{R}^n$$

$$\iff \mathrm{rank}(W_R(t_0,t_1)) = n \quad \text{or} \quad \mathrm{Im}(W_R(t_0,t_1)) = \mathbb{R}^n$$

For LTI systems:

$$\mathrm{Im}(W_R(t_0,t_1)) = \mathrm{Im}(\Gamma) \qquad \Gamma = [B \ AB \ \cdots \ A^{n-1}B]$$

$$R[t_0,t_1] = \mathrm{Im}(W_R(t_0,t_1)) = \mathrm{Im}(\Gamma) = \mathrm{Im}(W_C(t_0,t_1)) = C[t_0,t_1]$$

← Place eigenvalues Arbitrarily if Completely Controllable

Example 1

Is the Following Controllable?

$$A = \begin{bmatrix} -2 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\Gamma = [B \ AB \ A^2B]$$

To show this we can use any of the following

$$\mathrm{rank}(\Gamma) = n \iff \Gamma\Gamma^T \text{ non-singular} \iff \det(\Gamma\Gamma^T) \neq 0$$

Example 2

$$\dot{x} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \qquad \Gamma = [B \ AB] = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}$$

$\Rightarrow$ This system is not completely controllable

So what is the image of Γ

$$\mathrm{Im}(\Gamma) = \{ y \in \mathbb{R}^n \text{ s.t. } \exists\, x \in \mathbb{R}^n \text{ where } \Gamma x = y \}$$

$$\Gamma x = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + 3x_2 \\ x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \Rightarrow y_1 = y_2$$

$$\Rightarrow \mathrm{Im}(\Gamma) = \left\{ y \mid y = \begin{bmatrix} \alpha \\ \alpha \end{bmatrix} \text{ for } \alpha \in \mathbb{R} \right\}$$

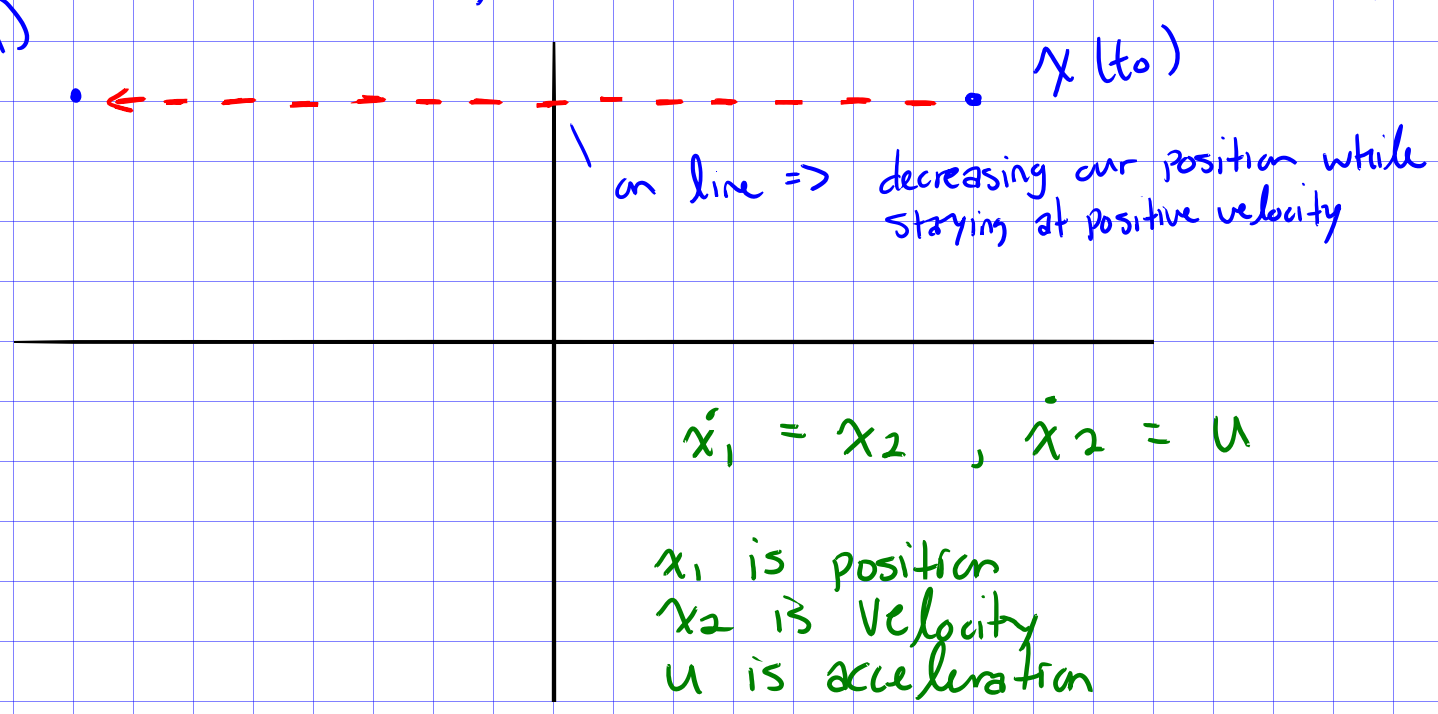
$\Rightarrow$ We can go from $\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ to 0 and from zero to $\beta \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Example 3

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \qquad \mathcal{M} = [B \;\; AB]$$

$$\mathcal{M} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \text{rank}(\mathcal{M}) = 2$$ complete controllability

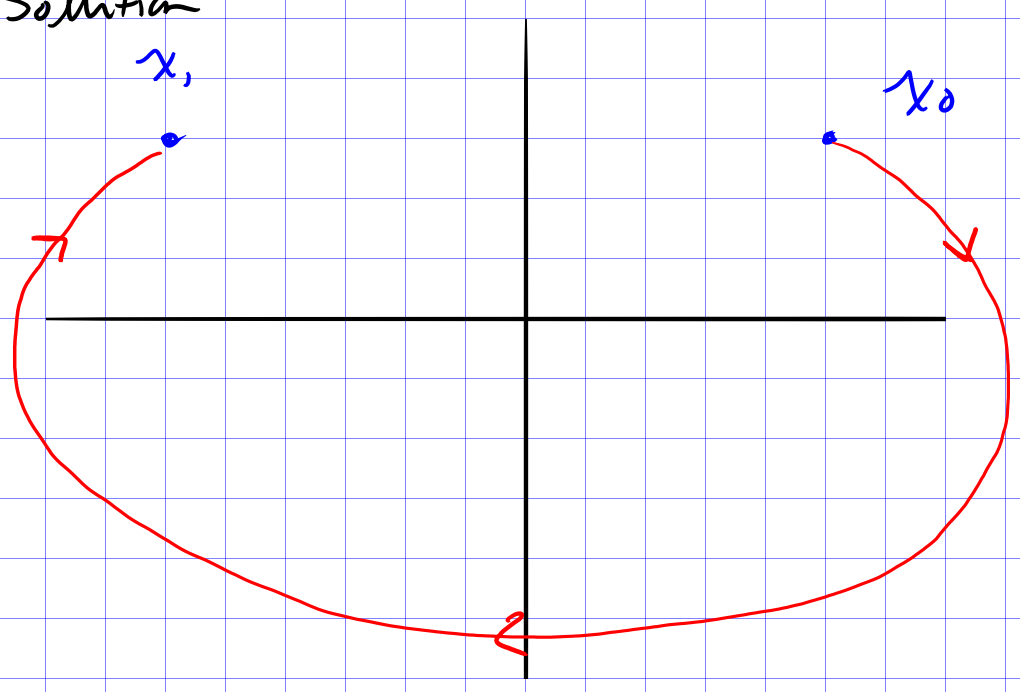
You can go between any two points, but can you follow any trajectory?



$\dot{x}_1 = x_2$, $\dot{x}_2 = u$

x_1 is position
x_2 is velocity
u is acceleration

Actual Solution



C.C. => You can get from any point to any other point, it does not imply that you can follow any trajectory.